BUILD A GREENER WORLD
CREATE A BETTER LIFE

Power Atlantic 5MWh Battery Container

Specification

(Premilitary Version)

PREPARED BY:

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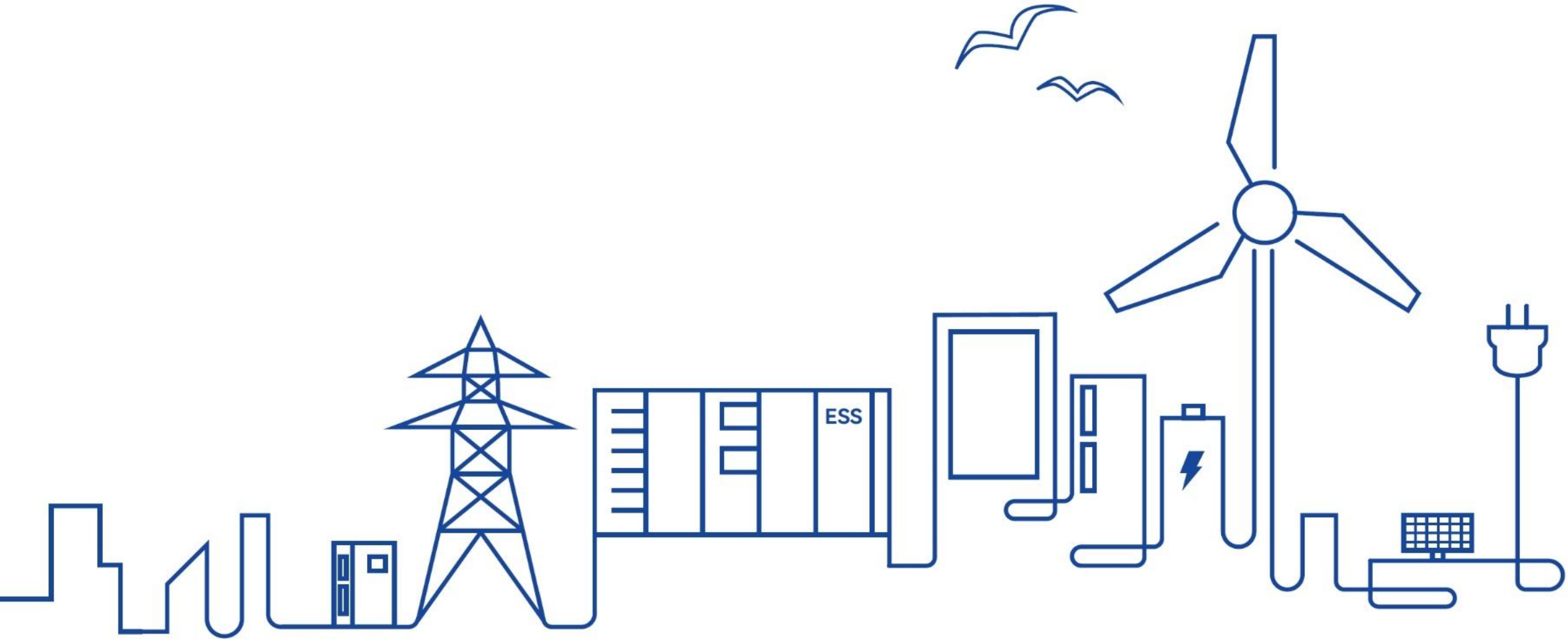

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1. Lithium Iron Phosphate Battery Container

1.1 Functions and Features

1.1.1 Product Functions

This container energy storage system is mainly used for storing and transferring energy, which supports battery status monitoring and management, wide application such as grid peak regulation, frequency regulation, grid-connected access of new energy power stations, etc.

1.1.2 Product Features

This Liquid cooling energy storage system is a highly integrated product, which integrates battery system, battery management system, thermal management system, fire suppression system, power supply and distribution system, monitoring system, environmental control system, and cable routing, etc., and has the characteristics of high safety and reliability, easy installation, high energy density and long cycle life.

(1) High energy density

The large module technology and 1500V high-voltage group technology are used to substantially improve the energy density of the energy storage system, reduce the material cost of single-system equipment and the land cost of the energy storage power station, and improve the economy of the system.

(2) High safety

High-sensitivity temperature, smoke and gas composite detection provide high-reliability protection with the concept of timely detection, timely response, early warning, and prevention of accident expansion. The double-layer fire extinguishing mechanism by aerosol + backup water is used to ensure product safety under extreme conditions.

1.2 Product Appearance

Figure 2-1 The dimension of the container system

1.3 System Description

1.3.1 System Specification

Table 2-1 Parameters of energy storage system

Item	Specification
Battery type	314Ah
Configuration	12P416S
Rated energy/ kWh	5015
Battery voltage range/ V	1164.8 – 1497.6
Duration/ h	≥2
Communication	Ethernet / CAN / RS485
Thermal Management	Liquid cooling

Dimension (W x D x H)/ mm		6,058 x 2,438 x 2,896
Weight/ T		~ 43
Operation temperature/ ℃	Charge	0 - 55
	Discharge	-20 - 55
Humidity (storage/operating)		＜95% RH (non-condensing)
IP level		IP54

1.3.2 System Architecture

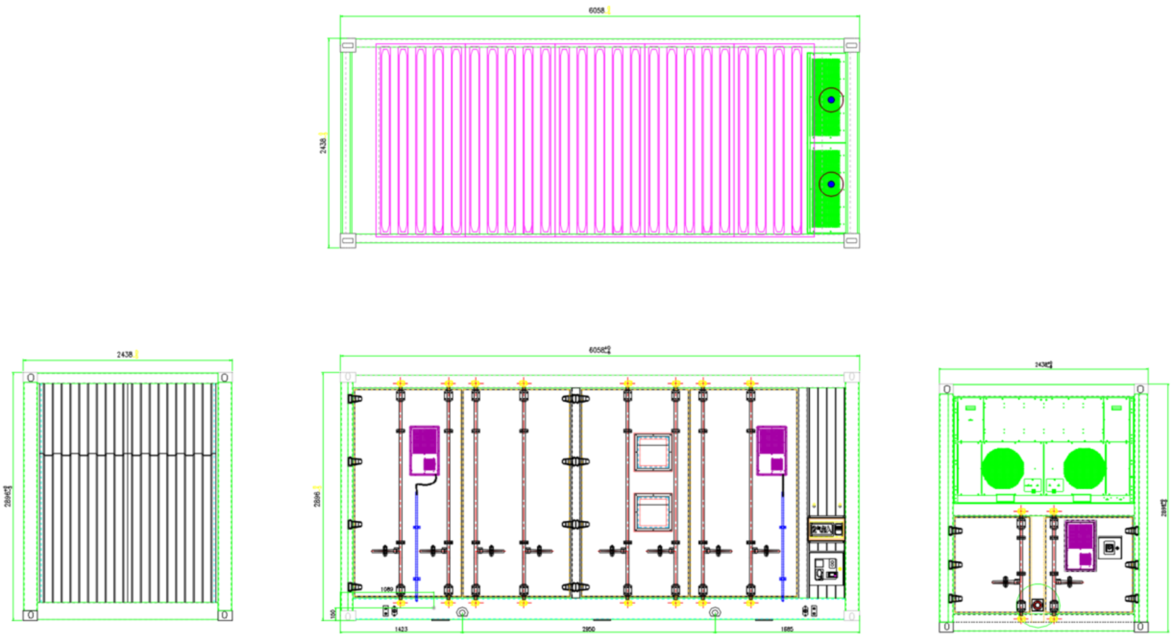

Figure 2-2 The structure diagram of the container system

1.3.3 System Components

Table 2-2 Components of the container system

No.	Sub-parts	Components	Quantity	Remark
1	Battery container	20HC	1	Including distribution and lighting system
2	Battery system	Battery pack	48	1P104S
		Battery cluster switchgear unit	12	Including BCMU, breaker, fuse, contactor, current sensor, etc.
		DC combiner cabinet	1	Including BAMS, disconnector, UPS
3	Liquid cooling system	Liquid cooling machine	1	Cooling capacity: 45kW

		Liquid cooling pipes	1	Three-stage pipes
4	Fire suppression system	Fire suppression controller	1	Including aerosol fire extinguishing reagent
		Smoke sensor	3	For smoke detection
		Flammable gas sensor	2	H_2
		CO sensor	2	CO
		Exhaust fan	1	
		Ventilation system	1	Flammable gas in battery compartment

1.4 Component Description

1.4.1 Battery Cell

Table 2-3 Parameters of battery cell

Item		**Specification**
Battery type		LFP
Capacity/ Ah		314
Nominal voltage/ V		3.2
Voltage range/ V		2.5 - 3.65
Rated energy/ Wh		1004.8
Charge/Discharge Current	Standard	0.5C
	Max.	1C
Energy density/ Wh/kg		183
Life cycle (25℃, 100%DOD, 80%SOH)		6,000 (0.5C/0.5C)
Dimension (W x D x H)/ mm		72×174×207.7
Weight/ Kg		5.5
Operation temperature/ ℃	Charge	0 - 55
	Discharge	-20 - 55

1.4.2 Battery Pack

Table 2-4 Parameters of battery pack

Item	**Specification**

Pack type		1P104S
Capacity/ Ah		314
Rated energy/ kWh		104.499
Configuration		1P104S
Nominal voltage/ V		332.8
Voltage range/ V		291.2 – 374.4
Duration/ h		≥2
Operation temperature/ °C	Charge	0 - 55
	Discharge	-20 - 55
Dimension (W x D x H)/ mm		762.5x 2180 x 252
Weight/ Kg		~690±10
IP level		IP67
Thermal management		Liquid cooling

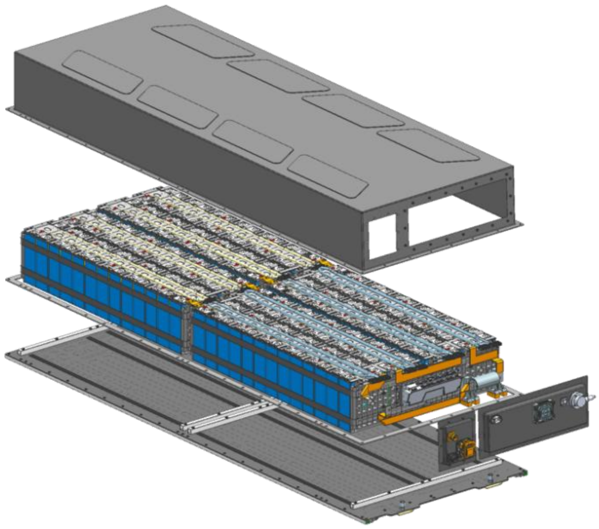

Figure 2-3 The diagram of battery pack (for reference only)

1.4.3 Battery Cluster

Table 2-5 Parameters of battery cluster

Item	Specification
Rack type	1P416S
Battery type	LFP
Capacity/ Ah	314
Rated energy/ kWh	417.996

Configuration		1P416S
Nominal voltage/ V		1331.2
Voltage range/ V		1164.8 – 1497.6
Duration/ h		≥2
Operation temperature/ °C	Charge	0 - 55
	Discharge	-20 - 55
Weight/ T		~ 2.7
IP level		IP54
Thermal management		Liquid cooling

1.4.4 Liquid Cooling System

The battery part of the system adopts liquid cooling technology to dissipate heat. Liquid cooling technology brings less loss and better temperature uniformity. The liquid cooling system mainly comprises of pipes, pumps, heat exchangers and compressors. The coolant of the system is mixed solution of ethylene glycol and water. The coolant flows from the water inlet pipe to the 10 main longitudinal branches. Each main branch is divided into 8 branches and flows to the liquid cooling pack to cool / heat the cell. After flowing out of the pack, it is summarized to the return pipe through the branch pipe and returned to the liquid cooling unit.

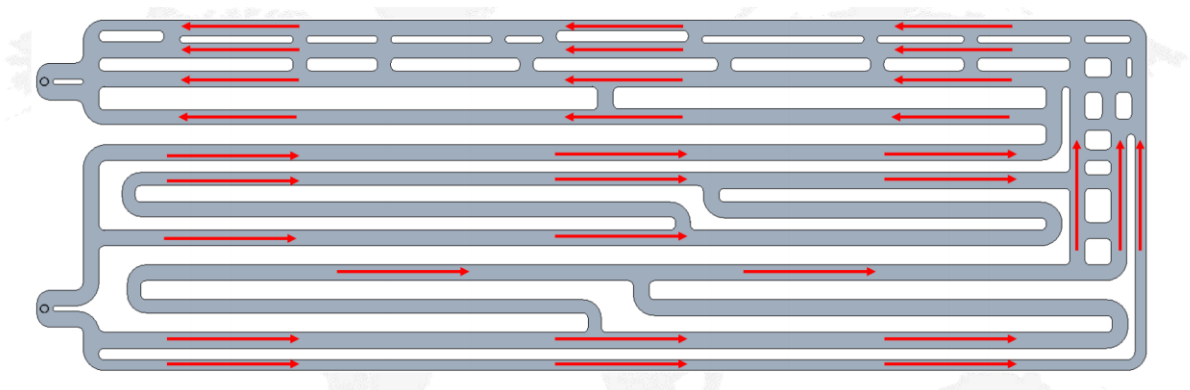

Figure 2-4 Flow direction of coolant in pack

The operating principal diagram of the liquid cooling unit is shown in the figure below.

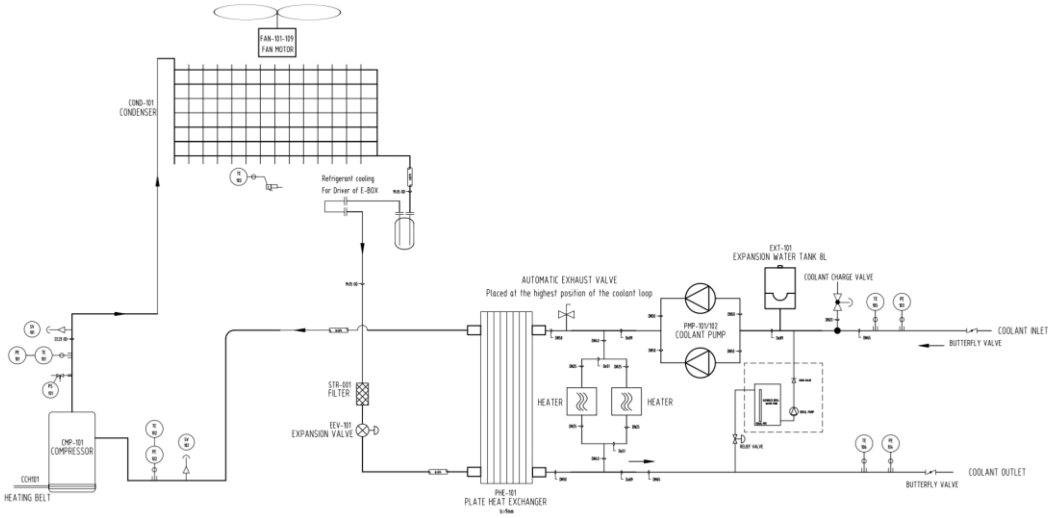

Figure 2-5 Operating principle diagram

Refrigerant Cycle: The compressor compresses the medium temperature and low-pressure gas from the evaporator into high temperature and high-pressure gas. When the gas pass through the condenser, they will liquefy and release heat into medium temperature and high-pressure liquid, and then pass through the electronic expansion valve to become a gas-liquid mixed state with low temperature and low pressure (the gas accounts for a small proportion), and finally enter the evaporator. After sufficient heat exchange in the evaporator, the refrigerant turns into a low-temperature and low-pressure gas, and then repeat the next cycle.

Coolant Cycle: The coolant is sent to the battery by the pump. After absorbing the heat generated by the battery it is returned to the evaporator to exchange heat, then repeat the next cycle.

1.4.5 Fire Suppression System

1）Fire protection system introduction

The battery liquid-cooled container system uses an aerosol fire suppression system and a water-based fire protection system. Each liquid-cooled container is equipped with smoke detectors, heat detectors, combustible gas detectors, and CO detectors, which communicate with the fire control mainframe. The container is also equipped with a fire control mainframe, audible and visual alarms, gas release indicators, and emergency start/stop buttons.

When any detector inside the compartment is triggered, the fire control mainframe receives a primary alarm, and the audible and visual alarm installed on the container will be activated.

When a smoke detector and a heat detector simultaneously trigger an alarm, the fire control mainframe will issue a secondary alarm, the outdoor audible and visual alarm will be activated, the gas fire suppression system will be initiated, and the extinguishing agent will be discharged into the battery compartment. At the same time, the gas release indicator will be activated. The compartment is also equipped with explosion-proof combustible gas detectors and CO detectors. When the detectors reach low or high thresholds, the explosion-proof fans will be activated. Once the concentration of combustible gases drops below the low threshold or the fire protection system reaches a secondary alarm level, the intake and exhaust fans can automatically shut down. The fire control mainframe can issue fire alarm and fault dry contact signals to the battery management system.

When the integrated device in the battery pack reaches a secondary or tertiary alarm level, the fire control mainframe will issue a primary external alarm, which can activate the audible and visual alarm. When it reaches a quaternary alarm level, the fire control mainframe will issue a secondary external alarm, activate the audible and visual alarm, and start a 30-second countdown. At the end of the countdown, the Aerosol system will be activated to spray the extinguishing agent.

2）Liquid cooling container fire suppression process

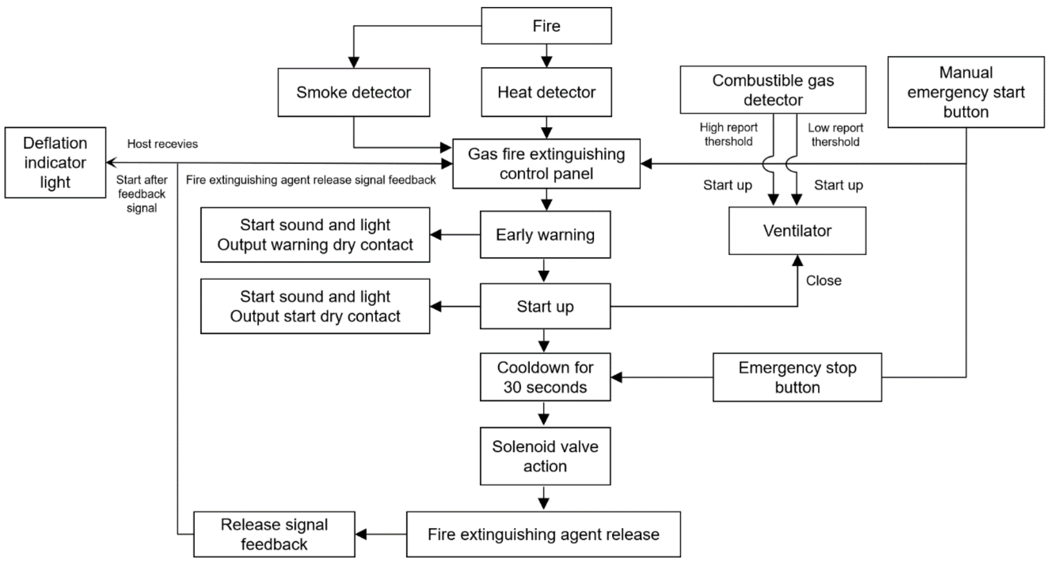

Figure 2-6 Fire alarm process diagram

1.4.6 Battery Management System (BMS)

The battery system uses three level BMS (Battery Management System). It consists of the battery management unit BMU, battery cluster management unit BCMU, and battery stack management unit BAMS. The main functions of BMS are described as follows:

(1) Operation control function: including start-stop control, charge and discharge control, operation parameter setting and thermal management control.

(2) Data acquisition function: BMS should be able to measure the battery's electrical and thermal related data in real time, including single battery voltage, battery module temperature, battery module voltage, series circuit current, insulation resistance, battery pole temperature, leakage monitoring and other parameters.

(3) Alarm protection function: BMS has the electrical protection function of battery overvoltage protection, undervoltage protection, overcurrent protection, short circuit protection, overtemperature protection, low temperature protection, leakage protection, etc., and can send alarm signals or trip instructions, implement local fault isolation, and report alarm information.

(4) Fault diagnosis function: The BMS should be able to monitor the operating status of the battery in real time, diagnose the abnormal operating status of the battery and the BMS itself, and upload the alarm signal to the local monitoring system and power conversion system.

(5) Operation management function: BMS can effectively manage charge and discharge to ensure that no overcharge or over discharge of the battery occurs in the process of charge and discharge, to prevent charging and discharging current and temperature from exceeding the allowable value.

Each battery box is configured with two battery management units (BMUs). The BMUs collect battery data such as voltage and temperature and communicate with the upper-layer main control module through the CAN. The battery cluster is equipped with a control box. The control box integrates the BMS secondary master BCMU and the battery cluster control circuit to manage and control the battery cluster, and upload the status information of the battery cluster to the upper management unit. The 18 clusters of batteries in the entire system are controlled and managed by two BMS system management hosts, which upload the status data of the battery system to the energy storage system monitoring device and receive the control instructions of the superior device. The battery system management host also provides display and storage functions.

1)BMU parameter

Table 2-6 BMU parameter specifications requirements

No.	Name of technical parameter	Unit	Parameter
1	Number of voltage samples	PCS	52
2	Temperature collection quantity	PCS	12
3	Cell voltage measurement range	V	0~5
4	Cell voltage detection accuracy	mV	±5
5	Cell voltage sampling period	ms	≤100ms
6	Cell temperature detection range	℃	-40~105℃
7	Cell temperature detection accuracy	℃	±1
8	Cell temperature sampling period	ms	≤200ms
9	Battery equalization mode	/	Passive equalization
10	Battery equalizing current	mA	≥80mA
11	BMU power supply	VDC	24
12	BMU power consumption	W	≤2
13	Communication mode	/	CAN2.0

2)BCMU parameter

Table 2-7 BCMU parameter specifications requirements

No.	Name of technical parameter	Unit	Parameter
1	Total voltage measurement range	V	0~1500
2	Total voltage measurement accuracy	%	±0.5%FSR
3	Total voltage measurement period	ms	≤100ms
4	Current measuring range	A	±400
5	Current measurement accuracy	%	±0.5%FSR
6	Current measurement period	ms	≤100ms
7	SOC calculation	/	≤5%
8	Electrical energy calculation error	/	≤±2%
9	Communication mode	/	CAN/RS485/Modbus
10	Communication interface	/	CAN/RS485
11	BCMU power supply	VDC	24
12	BCMU power consumption	W	≤3

3)BAMS parameter

Table 2-8 BAMS parameter specifications requirements

No.	Name of technical parameter	Unit	Parameter
1	Operating voltage system	VDC	24
2	Operating temperature range	℃	-20~+85
3	Maximum working humidity	%	20%~90%RH
4	Temperature detection range	℃	-20~+105
5	Operating power consumption	W	≤5
6	Communication mode	/	CAN/RS485/Modbus TCP/RTU
7	Communication interface	/	CAN/RS485, ethernet